

**Laxmi Charitable Trust's**  
**Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce**  
**Department of Information Technology (Bsc.IT Semester III)**  
**Python for Data Science Practical**

## Practical - II

|                               |                                    |
|-------------------------------|------------------------------------|
| Roll No. : 44                 | Name: Amit Umarvaish               |
| Class : SY IT                 | Batch : 1st                        |
| Date of Assignment : 22/07/26 | Date/Time of Submission : 22/07/26 |

2a. Write a python code to create nested tuples.

Code:

```
student = (("Name", "Amit"),("Age", 21),("City",  
"Mumbai"),("Course", "Python"))
```

```
print(student)
```

```
print("Name:", student[0][1])
```

```
print("Age:", student[1][1])
```

```
print("City:", student[2][1])
```

```
print("Course:", student[3][1])
```

```
print("Amit 44")
```

```
student = (("Name", "Amit"),("Age", 21),("City", "Mumbai"),("Course", "Python"))
```

```
print(student)
```

```
print("Name:", student[0][1])
```

```
print("Age:", student[1][1])
```

```
print("City:", student[2][1])
```

```
print("Course:", student[3][1])
```

```
print("Amit 44")
```

Output:

```
==== RESTART: C:/Users/11-4//AppData/Local/Programs/Python/P
(('Name', 'Raj'), ('Age', 20), ('City', 'Mumbai'), ('Course', 'Python'))
Name: Raj
Age: 20
City: Mumbai
Course: Python
Amit 44
```

2b. Write a python code to sort the nested tuple using sorted () function.

Code:

```
data = ("Amit","Raj","Jay")
sorted_data = sorted(data, key = lambda x:x[1])

print(sorted_data)

print("Amit 44")
```

---

```
data = ("Amit","Raj","Jay")
sorted_data = sorted(data, key = lambda x:x[1])

print(sorted_data)

print("Amit 44")
```

Output:

```
==== RESTART: C:/I
['Raj', 'Jay', 'Amit']
Amit 44
```

2c. Write a python code to copy or clone list.

Code:

```
list1 = ("Amit","Raj","Jay")
list2 = list1
```

**print(list2)**

**print("Amit 44")**

```
list1 = ("Amit","Raj","Jay")
list2 = list1

print(list2)
print("Amit 44")
```

Output:

```
==== RESTART: C:/I
('Amit', 'Raj', 'Jay')
Amit 44
```

2d. Write a python code to check immutability property of python tuples.

Code:

**name = ("Amit","Raj","Jay")**  
**print("Name",name)**

**name[1] = 30**

**print(name)**

```
name = ("Amit","Raj","Jay")
print("Name",name)

name[1] = 30
print(name)
```

Output:

```
==== RESTART: C:/Users/IT-47/AppData/Local/Programs/Python/Python38-64/Python.exe  
Name ('Amit', 'Raj', 'Jay')
```

```
Traceback (most recent call last):
```

```
File "C:/Users/IT-47/AppData/Local/Programs/Python/Python38-64/Python.exe", line 1, in <module>  
    name[1] = 30
```

```
TypeError: 'tuple' object does not support item assignment
```